

Recall and demonstrate proficiency with multiplication facts up to 10×10 and related division facts; extend and apply facts to develop efficient mental strategies for computation with larger numbers without a calculator

Learning intention

- recall and demonstrate proficiency with multiplication facts up to 10×10 and related division facts; extend and apply facts to develop efficient mental strategies for computation with larger numbers without a calculator.

Teach and model

- using arrays on grid paper or created with blocks or counters to develop.
- designing, creating and playing instructive card games that involve the recall.
- using materials or diagrams to develop and record multiplication strategies such as doubling, halving, commutativity.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.